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import numpy as np
import matplotlib.pyplot as plt
from collections import Counter

# STEP 1: CREATE SAMPLE DATA
# Features: [Hours Studied, Hours Slept]
X_train = np.array([
    [1, 1], [2, 2], [3, 1], [2, 3], # Fail group
    [6, 7], [7, 8], [8, 6], [9, 9], # Pass group
    [5, 5], [4, 6]                 # Borderline cases
])
# Labels: 0 = Fail, 1 = Pass
y_train = np.array([0, 0, 0, 0, 1, 1, 1, 1, 1, 0])

# STEP 2: THE KNN FUNCTION
def knn_predict(X_train, y_train, x_new, k):
    # 1. Calculate Euclidean Distance
    distances = []
    for i in range(len(X_train)):
        dist = np.sqrt(np.sum((x_new - X_train[i])**2))
        distances.append((dist, y_train[i]))

    # 2. Sort by distance and pick top K
    distances.sort(key=lambda x: x[0])
    neighbors = distances[:k]

    # 3. Majority Vote
    labels = [n[1] for n in neighbors]
    most_common = Counter(labels).most_common(1)
    return most_common[0][0]

# STEP 3: FINDING THE BEST K (Selection Logic)
k_range = [1, 3, 5, 7]
accuracies = []

for k in k_range:
    correct_predictions = 0
    # Simple "Leave-One-Out" validation
    for i in range(len(X_train)):
        # Temporarily remove one point to test the model
        test_point = X_train[i]
        true_label = y_train[i]

        X_temp = np.delete(X_train, i, axis=0)

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y_temp = np.delete(y_train, i)

prediction = knn_predict(X_temp, y_temp, test_point, k)
if prediction == true_label:
    correct_predictions += 1

accuracies.append(correct_predictions / len(X_train))

# Find the K with highest accuracy
best_k = k_range[np.argmax(accuracies)]
print(f"Best K value found: {best_k} with accuracy {max(accuracies)*100}%")

# STEP 4: VISUALIZATION
plt.figure(figsize=(12, 5))

# Plot 1: The Data Points
plt.subplot(1, 2, 1)
plt.scatter(X_train[y_train==0][:,0], X_train[y_train==0][:,1], color='red', label='Fail')
plt.scatter(X_train[y_train==1][:,0], X_train[y_train==1][:,1], color='green', label='Pass')
plt.title("Student Data Distribution")
plt.xlabel("Study Hours")
plt.ylabel("Sleep Hours")
plt.legend()

# Plot 2: Selection of K (Accuracy Plot)
plt.subplot(1, 2, 2)
plt.plot(k_range, accuracies, marker='o', linestyle='dashed', color='blue')
plt.title("Accuracy vs K Value")
plt.xlabel("K")
plt.ylabel("Accuracy Score")
plt.xticks(k_range)

plt.tight_layout()
plt.show()

# STEP 5: PREDICT FOR A NEW STUDENT
new_student = np.array([5, 4]) # 5 hours study, 4 hours sleep
final_result = knn_predict(X_train, y_train, new_student, k=best_k)
print(f"Prediction for new student {new_student}: {'Pass' if final_result == 1 else 'Fail'}")

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